Swiggy Sales Analysis

In [ ]:
```
# Import Libraries python
```

In [13]:
```
!pip install pandas numpy  seaborn plotly
```

```
Defaulting to user installation because normal site-packages is not writea
ble
Requirement already satisfied: pandas in c:\programdata\anaconda3\lib\site
-packages (2.3.3)
Requirement already satisfied: numpy in c:\programdata\anaconda3\lib\site-
packages (2.3.5)
Requirement already satisfied: seaborn in c:\programdata\anaconda3\lib\sit
e-packages (0.13.2)
Requirement already satisfied: plotly in c:\programdata\anaconda3\lib\site
-packages (6.3.0)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\programdata\an
aconda3\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\programdata\anaconda3\li
b\site-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\programdata\anaconda3
\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in c:\programdata\a
naconda3\lib\site-packages (from seaborn) (3.10.6)
Requirement already satisfied: narwhals>=1.15.1 in c:\programdata\anaconda
3\lib\site-packages (from plotly) (2.7.0)
Requirement already satisfied: packaging in c:\programdata\anaconda3\lib\s
ite-packages (from plotly) (25.0)
Requirement already satisfied: contourpy>=1.0.1 in c:\programdata\anaconda
3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\programdata\anaconda3\li
b\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\programdata\anacond
a3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\programdata\anacond
a3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.4.9)
Requirement already satisfied: pillow>=8 in c:\programdata\anaconda3\lib\s
ite-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (12.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\programdata\anaconda
3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.2.5)
Requirement already satisfied: six>=1.5 in c:\programdata\anaconda3\lib\si
te-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
```

In [14]:
```
!pip install matplotlib
```

```
Defaulting to user installation because normal site-packages is not writea
ble
Requirement already satisfied: matplotlib in c:\programdata\anaconda3\lib
\site-packages (3.10.6)
Requirement already satisfied: contourpy>=1.0.1 in c:\programdata\anaconda
3\lib\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\programdata\anaconda3\li
b\site-packages (from matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\programdata\anacond
a3\lib\site-packages (from matplotlib) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\programdata\anacond
a3\lib\site-packages (from matplotlib) (1.4.9)
Requirement already satisfied: numpy>=1.23 in c:\programdata\anaconda3\lib
\site-packages (from matplotlib) (2.3.5)
Requirement already satisfied: packaging>=20.0 in c:\programdata\anaconda3
\lib\site-packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in c:\programdata\anaconda3\lib\s
ite-packages (from matplotlib) (12.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\programdata\anaconda
3\lib\site-packages (from matplotlib) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in c:\programdata\anac
onda3\lib\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in c:\programdata\anaconda3\lib\si
te-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
```

In [1]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

print("All libraries imported successfully!")
```

```
All libraries imported successfully!
```

Import Data

In [7]:

```
import pandas as pd

df = pd.read_csv("C:/Users/hemal/Downloads/swiggy_data_python.csv")

df
```

Out[7]:

	State	City	Order Date	Restaurant Name	Location	Category	
0	Karnataka	Bengaluru	29-06-2025	Anand Sweets & Savouries	Rajarajeshwari Nagar	Snack	Mu
1	Karnataka	Bengaluru	03-04-2025	Srinidhi Sagar Deluxe	Kengeri	Recommended	
2	Karnataka	Bengaluru	15-01-2025	Srinidhi Sagar Deluxe	Kengeri	Recommended	
3	Karnataka	Bengaluru	17-04-2025	Srinidhi Sagar Deluxe	Kengeri	Recommended	
4	Karnataka	Bengaluru	13-03-2025	Srinidhi Sagar Deluxe	Kengeri	Recommended	
...	...	...	...	...	...	...	
197425	Sikkim	Gangtok	25-01-2025	Mama's Kitchen	Gangtok	Momos	m
197426	Sikkim	Gangtok	02-07-2025	Mama's Kitchen	Gangtok	Momos	K
197427	Sikkim	Gangtok	25-03-2025	Mama's Kitchen	Gangtok	Momos	
197428	Sikkim	Gangtok	26-03-2025	Mama's Kitchen	Gangtok	Momos	I
197429	Sikkim	Gangtok	27-03-2025	Mama's Kitchen	Gangtok	Momos	

197430 rows × 10 columns

EDA

In [28]:

```
df.head()
```

Out[28]:

	State	City	Order Date	Restaurant Name	Location	Category	Dish Name
0	Karnataka	Bengaluru	29-06-2025	Anand Sweets & Savouries	Rajarajeshwari Nagar	Snack	Butte Murukku 200gn
1	Karnataka	Bengaluru	03-04-2025	Srinidhi Sagar Deluxe	Kengeri	Recommended	Badan Mil
2	Karnataka	Bengaluru	15-01-2025	Srinidhi Sagar Deluxe	Kengeri	Recommended	Chov Chov Batl
3	Karnataka	Bengaluru	17-04-2025	Srinidhi Sagar Deluxe	Kengeri	Recommended	Kesar Batl
4	Karnataka	Bengaluru	13-03-2025	Srinidhi Sagar Deluxe	Kengeri	Recommended	Mi: Raith

In [29]:

```
df.tail()
```

Out[29]:

	State	City	Order Date	Restaurant Name	Location	Category	Dish Name	Price (INR)	R
197425	Sikkim	Gangtok	25-01-2025	Mama's Kitchen	Gangtok	Momos	Soya cheese chilli momo ...	112.0	
197426	Sikkim	Gangtok	02-07-2025	Mama's Kitchen	Gangtok	Momos	Kurkure momo fried ...	140.0	
197427	Sikkim	Gangtok	25-03-2025	Mama's Kitchen	Gangtok	Momos	Chilli cheese momo	126.0	
197428	Sikkim	Gangtok	26-03-2025	Mama's Kitchen	Gangtok	Momos	Veg Momos (8 Pc)	85.0	
197429	Sikkim	Gangtok	27-03-2025	Mama's Kitchen	Gangtok	Momos	Soya Momo	100.0	

In [30]:

```
df.sample()
```

Out[30]:

	State	City	Order Date	Restaurant Name	Location	Category	Dish Name	Price (INR)
92743	West Bengal	Kolkata	19-03-2025	Burger King	Barasat	NEW Whopper Deluxe (Reg. Size Bun)	Extra Crunchy Veg Whopper Deluxe(Reg. Size Bun)!	149.0

In [31]:
```
df.describe()
```

Out[31]:

	Price (INR)	Rating	Rating Count
count	197430.000000	197430.000000	197430.000000
mean	268.512920	4.341582	28.321805
std	219.338363	0.422585	87.542593
min	0.950000	1.500000	0.000000
25%	139.000000	4.300000	0.000000
50%	229.000000	4.400000	2.000000
75%	329.000000	4.500000	15.000000
max	8000.000000	5.000000	999.000000

In [32]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 197430 entries, 0 to 197429
Data columns (total 10 columns):
 #   Column           Non-Null Count   Dtype  
---  ------           --------------   -----  
 0   State            197430 non-null  object 
 1   City             197430 non-null  object 
 2   Order Date       197430 non-null  object 
 3   Restaurant Name  197430 non-null  object 
 4   Location         197430 non-null  object 
 5   Category         197430 non-null  object 
 6   Dish Name        197430 non-null  object 
 7   Price (INR)      197430 non-null  float64
 8   Rating           197430 non-null  float64
 9   Rating Count     197430 non-null  int64  
dtypes: float64(2), int64(1), object(7)
memory usage: 15.1+ MB
```

In [35]:
```
df.loc[0:10]
```

Out[35]:

	State	City	Order Date	Restaurant Name	Location	Category	Di Nan
0	Karnataka	Bengaluru	29-06-2025	Anand Sweets & Savouries	Rajarajeshwari Nagar	Snack	Butt Murukk 200g
1	Karnataka	Bengaluru	03-04-2025	Srinidhi Sagar Deluxe	Kengeri	Recommended	Bada M
2	Karnataka	Bengaluru	15-01-2025	Srinidhi Sagar Deluxe	Kengeri	Recommended	Chc Chc Ba
3	Karnataka	Bengaluru	17-04-2025	Srinidhi Sagar Deluxe	Kengeri	Recommended	Kesa Ba
4	Karnataka	Bengaluru	13-03-2025	Srinidhi Sagar Deluxe	Kengeri	Recommended	M Raitl
5	Karnataka	Bengaluru	08-07-2025	Srinidhi Sagar Deluxe	Kengeri	Recommended	Srinid Sag Spec
6	Karnataka	Bengaluru	21-01-2025	Srinidhi Sagar Deluxe	Kengeri	Recommended	Gar Na
7	Karnataka	Bengaluru	13-04-2025	Srinidhi Sagar Deluxe	Kengeri	Recommended	Pis
8	Karnataka	Bengaluru	02-05-2025	Srinidhi Sagar Deluxe	Kengeri	North Indian Gravy	Panne Butt Masa
9	Karnataka	Bengaluru	30-07-2025	Srinidhi Sagar Deluxe	Kengeri	North Indian Gravy	Dal Tad
10	Karnataka	Bengaluru	03-08-2025	Srinidhi Sagar Deluxe	Kengeri	North Indian Gravy	Ka Mas

Metadata

In [38]:
```
df.shape
```

Out[38]: (197430, 10)

In [39]:
```
print("No of Rows:",df.shape[0])
```

No of Rows: 197430

In [40]:
```
print("No of Fields:",df.shape[1])
```

```
No of Fields: 10
```

In [41]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 197430 entries, 0 to 197429
Data columns (total 10 columns):
 #   Column           Non-Null Count   Dtype  
---  ------           --------------   -----  
 0   State            197430 non-null  object 
 1   City             197430 non-null  object 
 2   Order Date       197430 non-null  object 
 3   Restaurant Name  197430 non-null  object 
 4   Location         197430 non-null  object 
 5   Category         197430 non-null  object 
 6   Dish Name        197430 non-null  object 
 7   Price (INR)      197430 non-null  float64
 8   Rating           197430 non-null  float64
 9   Rating Count     197430 non-null  int64  
dtypes: float64(2), int64(1), object(7)
memory usage: 15.1+ MB
```

Data Types

In [43]:
```
df.dtypes
```

Out[43]:
```
State               object
City                object
Order Date          object
Restaurant Name     object
Location            object
Category            object
Dish Name           object
Price (INR)        float64
Rating             float64
Rating Count         int64
dtype: object
```

In [45]:
```
df.describe()
```

Out[45]:

	Price (INR)	Rating	Rating Count
count	197430.000000	197430.000000	197430.000000
mean	268.512920	4.341582	28.321805
std	219.338363	0.422585	87.542593
min	0.950000	1.500000	0.000000
25%	139.000000	4.300000	0.000000
50%	229.000000	4.400000	2.000000
75%	329.000000	4.500000	15.000000
max	8000.000000	5.000000	999.000000

KPI's

Total Sales

In [49]:
```
Total_sales = df["Price (INR)"].sum()
print("Total Sales (INR):",round(Total_sales,2))
```

```
Total Sales (INR): 53012505.77
```

Average Rating

In [53]:
```
Average_rating = df["Rating"].mean()
print("Average rating :",round(Average_rating,1))
```

```
Average rating : 4.3
```

Average Order value

In [56]:
```
Average_order_value = df["Price (INR)"].mean()
print("Avg Order value(INR):",round(Average_order_value,2))
```

```
Avg Order value(INR): 268.51
```

Ratings Count

In [57]:
```
rating_count = df["Rating Count"].sum()
print("Rating Count:",round(rating_count,2))
```

```
Rating Count: 5591574
```

Total orders

In [64]:
```
Total_orders = df["Order Date"].count()
print("Total_orders:",round(Total_orders,1))
```

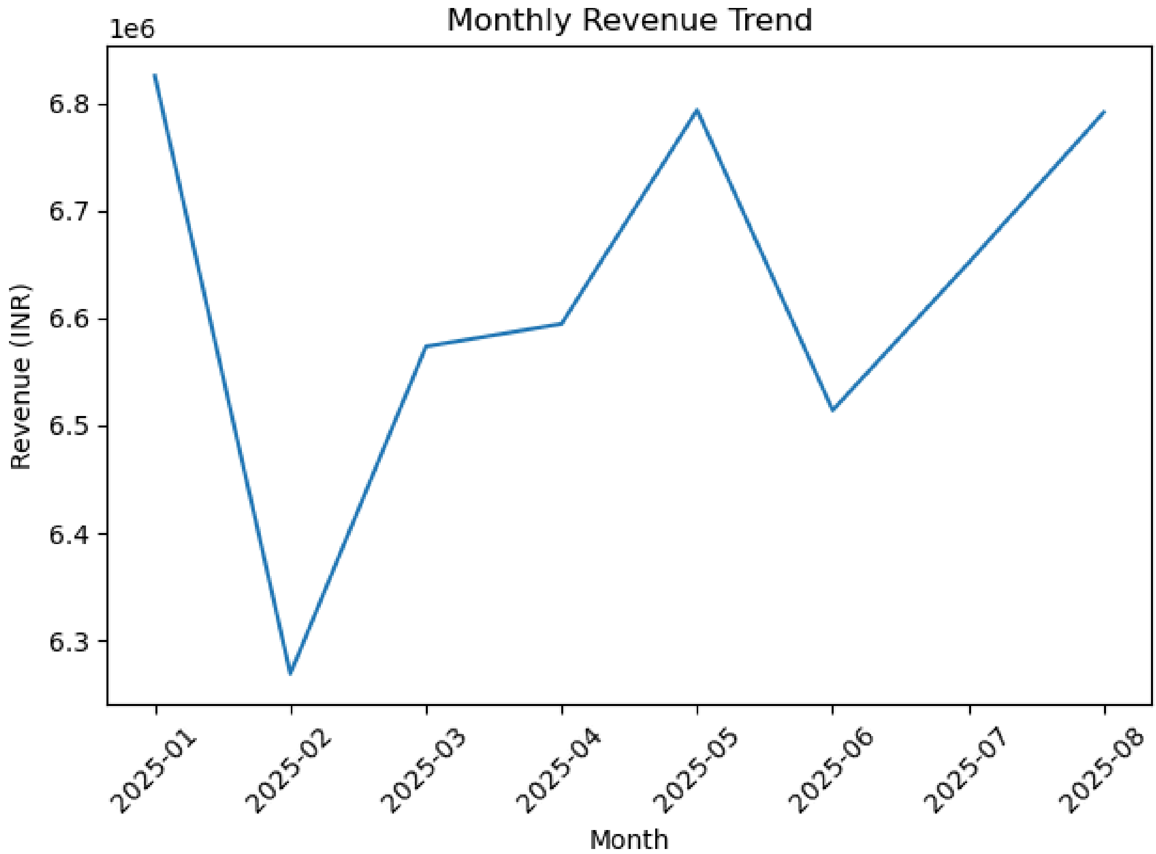
```
Total_orders: 197430
```

CHARTS Design

Monthly sales trend

In [67]:
```
df["Order Date"] = pd.to_datetime(df["Order Date"])
df["YearMonth"] = df["Order Date"].dt.to_period("M").astype(str)
monthly_revenue = df.groupby("YearMonth")["Price (INR)"].sum().reset_inde

plt.figure()
plt.plot(monthly_revenue["YearMonth"],monthly_revenue["Price (INR)"])
plt.xticks(rotation = 45)
plt.xlabel("Month")
plt.ylabel("Revenue (INR)")
plt.title("Monthly Revenue Trend")
plt.tight_layout()
plt.show()
```



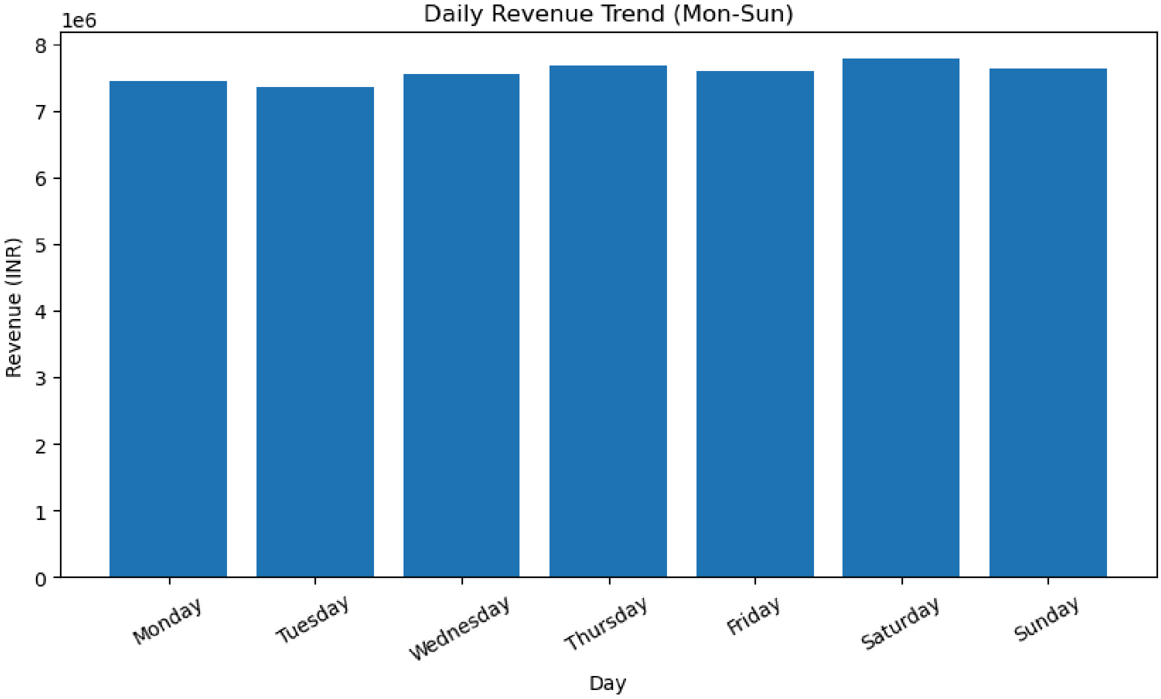
Daily Sales trend

In [69]:
```
df["DayName"] = pd.to_datetime(df["Order Date"]).dt.day_name()

daily_revenue = (
    df.groupby("DayName")["Price (INR)"]
      .sum()
      .reindex(["Monday","Tuesday","Wednesday","Thursday","Friday","Satur
)
plt.figure(figsize = (10,5))
plt.bar(daily_revenue.index, daily_revenue.values)
```

```
plt.title("Daily Revenue Trend (Mon-Sun)")
plt.xlabel("Day")
plt.ylabel("Revenue (INR)")
plt.xticks(rotation = 30)

plt.show()
```



Total Sales by food type

In [8]:
```
import numpy as np

non_veg_keywords = [
    "chicken",
    "egg",
    "fish",
    "mutton",
    "prawn",
    "biryani",
    "kabad",
    "kebab",
    "non-veg",
    "non veg"
]

df["Food Category"] = np.where(
    df["Dish Name"].str.lower().str.contains(
        "|".join(non_veg_keywords),
        na=False
    ),
    "Non-Veg",
    "Veg"
)
```

In [9]:
```
food_revenue = (
    df.groupby("Food Category")["Price (INR)"]
      .sum()
```

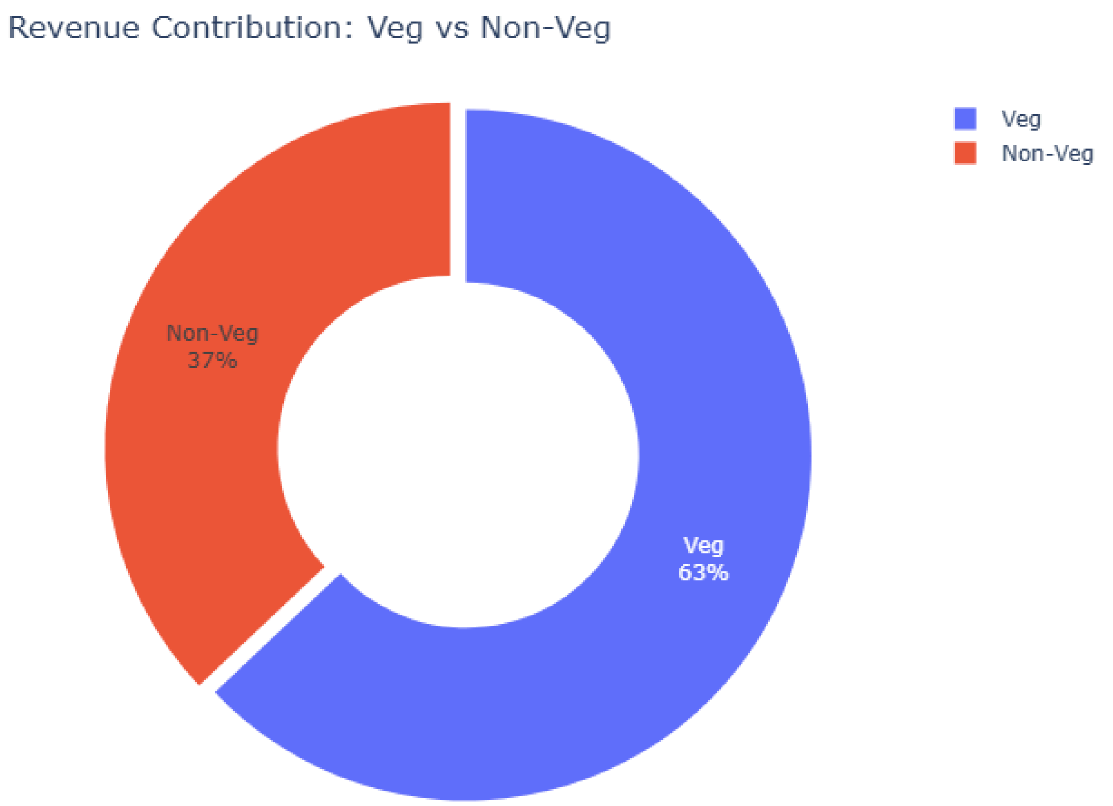
```
        .reset_index()
)
```

In [10]:
```
fig = px.pie(
    food_revenue,
    values="Price (INR)",
    names="Food Category",
    hole=0.5,
    title="Revenue Contribution: Veg vs Non-Veg",
)

fig.update_traces(
    textinfo="percent+label",
    pull=[0.05, 0]
)

fig.update_layout(
    height=500,
    margin=dict(t=60, b=40, l=40, r=40)
)

fig.show()
```



Total Sales by State

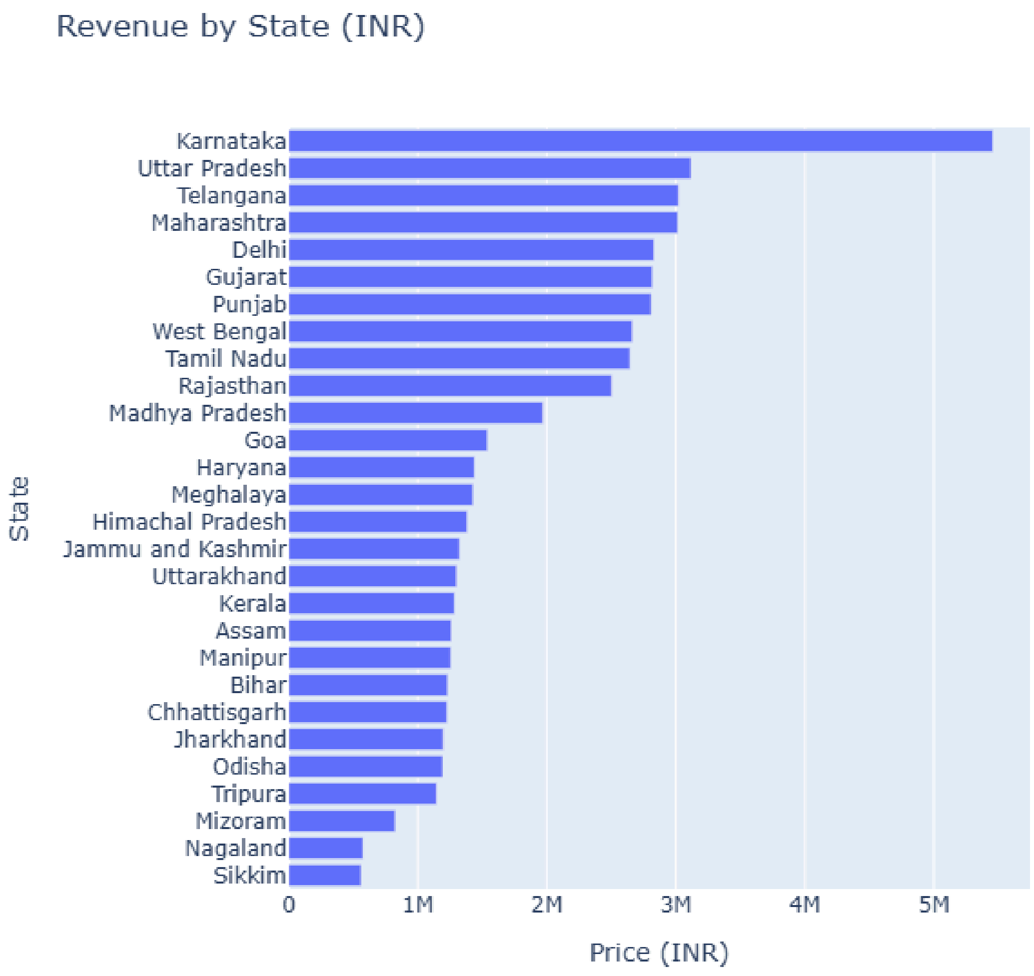
In [18]:
```
import plotly.express as px

fig = px.bar(
    df.groupby("State", as_index=False)["Price (INR)"].sum()
      .sort_values("Price (INR)", ascending=False),
    x="Price (INR)",
```

```
    y="State",
    orientation="h",
    title="Revenue by State (INR)"
)

fig.update_layout(
    height=600,
    yaxis=dict(autorange="reversed")
)

fig.show()
```



Quarterly Performance Summary

In [22]:
```
df["Order Date"] = pd.to_datetime(df["Order Date"])

df["Quarter"] = df["Order Date"].dt.to_period("Q").astype(str)

quarterly_summary = (
    df.groupby("Quarter", as_index=False)
      .agg(
          Total_Sales=("Price (INR)", "sum"),
          Avg_Rating=("Rating", "mean"),
          Total_Orders=("Order Date", "count")
      )
      .sort_values("Quarter")
```

```
)

quarterly_summary["Total_Sales"] = quarterly_summary["Total_Sales"].round
quarterly_summary["Avg_Rating"] = quarterly_summary["Avg_Rating"].round(2

quarterly_summary
```

Out[22]:

	Quarter	Total_Sales	Avg_Rating	Total_Orders
0	2025Q1	19667822.0	4.34	73096
1	2025Q2	19902257.0	4.34	74163
2	2025Q3	13442427.0	4.34	50171

Top 5 cities by sales

In [23]:

```
top_5_cities = (
    df.groupby("City")["Price (INR)"]
      .sum()
      .nlargest(5)
      .sort_values()
      .reset_index()
)

fig = px.bar(
    top_5_cities,
    x="Price (INR)",
    y="City",
    orientation="h",
    title="Top 5 Cities by Sales (INR)",
    color_discrete_sequence=["red"]
)

fig.show()
```